

PhD Research Proposal

Title

Abstract

1. Introduction

2. Problem Statement

3. Research Aim

4. Research Objectives

5. Research Questions

RQ1:

RQ2:

RQ3:

RQ4:

6. Literature Review

7. Research Gap

Analysis of recent studies indicates several limitations:

8. Proposed Methodology

Phase 1: Literature Survey

Comprehensive review

Phase 2: Dataset Acquisition

Collection of datasets from publicly available repositories and clinical collaborators.

Phase 3: Preprocessing

- Noise reduction
- Normalization

- Image augmentation

Phase 4: Deep Learning Model Development

- CNN-based architecture
- Transfer learning approaches
- Comparative model analysis

Phase 5: Explainability Integration

- Grad-CAM
- SHAP
- Feature importance analysis

Phase 6: Evaluation

Performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

9. Expected Outcomes

10. Significance of the Study

The proposed research addresses the critical need for interpretable AI solutions in healthcare. The framework is expected to improve diagnostic efficiency while maintaining transparency required for clinical decision-making.

11. Work Plan

Year 1:

- Literature review
- Research gap identification
- Dataset collection

Year 2:

- Model development
- Explainability integration
- Experimental evaluation

Year 3:

- Validation
- Journal publications
- Thesis writing

12. Expected Publications

- One Scopus-indexed review paper.
- Two Scopus-indexed research articles.
- Conference publications related to ACL diagnosis and explainable AI.

References

(To be completed from reviewed literature.)